

Gauge Structure from Boundary-Trace Fixation

Presentation Equivalence and the Minimal Gauge Role of Physical Carrier Structure

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Abstract

This paper derives the minimal gauge role within the realized-physics arc. Previous results established that realized persistence cannot be physically null, that standing-bearing persistence localizes, that localized standing induces continuation-neighborhood domains, that stable comparison supplies the metric role of a quotient comparison carrier, that perturbation transport is bounded by continuation-comparison constraints, that fields are distributed persistence carriers, and that particles are localized persistence classes. The present paper asks how multiple local presentations of the same carrier-mediated particle-role content can remain physically equivalent without becoming arbitrary representative choice. It proves that admissibility requires a boundary-trace-fixed presentation-equivalence structure: local presentations may vary only when carrier mediation, particle persistence-class identity, quotient comparison, boundary trace, and bounded transport are preserved.

The main result is a gauge-role induction theorem. In a fixed admissible realized persistence regime with field-carrier assignment Φ_D , particle persistence classes Π_D , bounded perturbation transport, quotient comparison carrier C_D , and boundary-trace fixation, presentation multiplicity that preserves the same standing-bearing physical role induces a gauge-role structure

$$\mathcal{G}_D = (\text{Pres}_D, \sim_G, \{g_{UV}\}, \text{Fix}_{\partial, D}).$$

This is the minimal physical gauge role: admissible local presentation equivalence and transition compatibility preserving the operative carrier and particle content already derived. Gauge equivalence is therefore not merely descriptive equivalence among notations; it is the boundary-fixed preservation of the same carrier-mediated transport/restoration organization across local presentations. Lie groups, principal bundles, fiber bundles, cocycle data, connections, curvature, Yang–Mills dynamics, gauge bosons, gauge couplings, BRST structure, and quantized gauge fields are not assumed. They are later specializations of the boundary-fixed presentation-equivalence role isolated here.

1 Introduction

At this stage of the realized-physics arc, familiar physical roles are reconstructed as consequences of admissible persistence structure. The first paper establishes the non-null floor of physical realization: realized persistence cannot be reconstruction-null inside its own admissible continuation regime [1]. The second derives bounded standing-localization as the origin of physical locality [2]. The third derives continuation-neighborhood domains as the origin of local-domain organization [3]. The fourth derives the metric role as quotient-stable comparative representation among local continuation domains [4]. The fifth derives bounded perturbation transport as the finite-propagation constraint role [5]. The sixth derives fields as persistence carriers [6]. The seventh derives particles as localized persistence classes [7].

The cumulative chain is

$$\text{RealPers}_D \Rightarrow \text{NonVacRec}_D \Rightarrow [B_D]_{\equiv_B} \Rightarrow (\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D) \Rightarrow C_D \Rightarrow \text{BoundTrans}_D \Rightarrow \Phi_D \Rightarrow \Pi_D.$$

The present paper begins at the final two terms. Once fields are understood as persistence-carrier mediation and particles as localized persistence classes, the next physical question is:

If the same carrier-mediated particle content can be presented locally in more than one admissible way, what prevents presentation freedom from becoming either false physical multiplicity or arbitrary convention?

The answer is gauge structure at its minimal role level. Gauge structure first enters realized physics as boundary-trace-fixed presentation equivalence preserving field-carrier mediation and particle persistence-class identity.

1.1 Theorem claim and physical role

The theorem-level claim is

$$(\Phi_D, \Pi_D, \text{BoundTrans}_D, C_D, \text{BoundaryFix}_D) \wedge \text{PresMult}_D \wedge \text{RolePres}_D \Rightarrow \text{GaugeRole}_D.$$

The physical-role claim is that GaugeRole_D is the minimal gauge role occupied by later formal gauge theories: it is the admissible equivalence-and-transition discipline by which locally variable presentations preserve the same carrier-mediated physical content.

This is a role-first derivation of gauge structure. The paper does not assume a Lie group, connection, curvature, covariant derivative, principal bundle, Yang–Mills action, gauge boson, coupling constant, or quantum gauge field. Rather, it derives the prior structural role those familiar objects later specialize: local presentation freedom constrained by boundary-trace-fixed preservation of physical content.

Paper	Structural result	Physical role
1	NonVacRec_D	physical realization cannot be null
2	$[B_D]_{\equiv_B}$	origin of physical locality
3	$(\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D)$	origin of local-domain organization
4	$C_D = \text{CompWit}_D / \sim_D$	origin of metric comparison role
5	BoundTrans_D	origin of finite-propagation constraint
6	Φ_D	origin of field-carrier mediation
7	$\Pi_D = [\kappa_U]_{\equiv_\Pi}$	origin of particle identity
8	$\mathcal{G}_D$	origin of gauge structure

1.2 Scope

The paper derives the minimal gauge role. It does not derive $U(1)$, $SU(2)$, $SU(3)$, Lie-algebra structure, connection one-forms, curvature two-forms, covariant derivatives, Yang–Mills equations, gauge bosons, gauge coupling values, BRST ghosts, anomaly cancellation, principal bundles, fiber bundles, cocycle structure, or the Standard Model gauge group. Those are later specializations. What is derived here is the prior physical role: multiple admissible presentations of the same carrier/particle content must be related by boundary-fixed equivalence and local transition maps that preserve standing-bearing content.

2 Dependencies and imported structure

The present argument depends on the first seven papers and on the boundary-trace and fixed-domain discipline developed in prior AASC work. The dependencies are summarized in Table 1.

Table 1: Imported structure for the gauge-role derivation.

Imported result	Role in the present paper
Non-vacuous realized persistence	Gauge cannot act on null content; presentation equivalence must preserve non-null physical role.
Bounded standing-localization	Gauge presentations must preserve bounded standing-localization, not relabel it away.
Continuation-neighborhood domains	Gauge transitions are local-domain transitions over overlaps or adjacency, not global defaults.
Quotient comparison carrier C_D	Presentation equivalence must preserve quotient-stable comparison classes.
Bounded perturbation transport	Gauge transitions must preserve admissible transport witnesses and bounded transport organization.
Field-carrier mediation Φ_D	Gauge acts on carrier presentations, not primitive field substance.
Particle persistence classes Π_D	Gauge equivalence must preserve particle-role identity rather than multiply it.
Boundary-trace fixation	Representative freedom is admissible only when boundary traces fix the role anchor.
AMetric boundary discipline	No default representative, hidden gauge selector, or primitive fiber is imported.
Anchor-tensor-skin discipline	Gauge role is tensor/role-level; notation-only transformations are skin.

The boundary-trace fixation results supply the central discipline. A local presentation may vary, but variation is not free-floating. It must be fixed by admissible boundary traces if it is to preserve the same physical role rather than import a selector or erase physical distinction. The gauge-coupling and Standard Model classification sources supply a physics-facing interpretation: standard gauge formalisms instantiate presentation equivalence, local transition compatibility, representation preservation, and invariant normalization, but those stronger structures are not assumed here.

3 Definitions

Definition 3.1 (Field-carrier assignment). The field-carrier assignment is the distributed carrier role derived previously,

$$\Phi_D : \text{Dom}(\Phi_D) \subseteq \text{LCD}_D \rightarrow \text{CarProf}_D.$$

For each local continuation domain U , $\Phi_D(U)$ is the carrier profile mediating standing-bearing persistence under bounded perturbation transport. The assignment is field-role structure, not primitive field substance.

Definition 3.2 (Particle persistence class). A particle persistence class is a stable localized carrier-mediated identity class

$$\Pi_D = [\kappa_U]_{\equiv_\Pi},$$

where κ_U is a localized carrier configuration and $\equiv_\Pi$ preserves standing signature, boundary/classification witness, restoration profile, quotient comparison, identity-memory, boundary trace, and admitted transport role.

Definition 3.3 (Local carrier presentation). For a local continuation domain $U \in \text{LCD}_D$, a local carrier presentation is an admissible representation

$$\rho_U \in \text{Pres}_D(U)$$

of the carrier/particle-role content over U . The presentation is not the content itself. It is also not merely a coordinate chart, notation choice, basis label, or field-variable name. It is an admissible local realization of the same carrier/particle-role structure: a local way of presenting the content carried by $\Phi_D(U)$ and, when present, the particle class Π_D supported there.

Definition 3.4 (Presentation multiplicity). $\text{PresMult}_D(U)$ holds when more than one admissible presentation represents the same carrier/particle role over U :

$$\rho_U, \rho'_U \in \text{Pres}_D(U), \quad \rho_U \neq \rho'_U, \quad \text{SameRole}_D(\rho_U, \rho'_U).$$

Presentation multiplicity is not physical multiplicity unless a boundary/classification witness distinguishes the presentations as different standing-bearing content.

Definition 3.5 (Same-role presentation equivalence). Write

$$\rho_U \sim_G \rho'_U$$

iff ρ_U and ρ'_U preserve the same standing-bearing carrier/particle-role content under boundary-trace-fixed redescription. Concretely, the equivalence preserves the relevant data

$$(\Phi, \Pi, S, I, R, P, Q, \partial, T, \chi, \beta),$$

namely carrier mediation, particle persistence-class identity, standing status, identity-memory, restoration status, perturbation burden, quotient comparison, boundary trace, transport interface, boundary/classification witness, and transport/restoration profile.

Definition 3.6 (Gauge transition). For overlapping or adjacent local continuation domains U, V , a gauge transition

$$g_{UV} : \rho_U \rightsquigarrow \rho_V$$

is an admissible boundary-compatible transition between local presentations preserving the same carrier/particle role across the overlap or adjacency. It is defined only where the relevant boundary traces are fixed and the transition preserves carrier mediation, particle persistence-class identity, quotient comparison, and bounded transport role.

Definition 3.7 (Gauge-role structure). The minimal gauge-role structure of D is

$$\mathcal{G}_D = (\text{Pres}_D, \sim_G, \{g_{UV}\}, \text{Fix}_{\partial, D}),$$

where Pres_D is the family of admissible local presentations, $\sim_G$ is same-role presentation equivalence, $\{g_{UV}\}$ is the family of boundary-compatible local presentation transitions, and $\text{Fix}_{\partial, D}$ is the boundary-trace fixation discipline anchoring those equivalences and transitions.

4 Presentation multiplicity and physical content

Gauge structure begins neither with a group nor with arbitrary convention. It begins with a distinction between variation in presentation and variation in physical content.

Proposition 4.1 (Presentation multiplicity does not imply physical multiplicity). *If ρ_U and ρ'_U differ only by admissible redescription preserving carrier mediation and particle persistence-class identity, then treating them as physically distinct duplicates standing-bearing content without a new boundary/classification witness.*

Proof. Physical distinction in the present framework requires a boundary-relevant difference in standing-bearing role. If two presentations preserve $(\Phi, \Pi, S, I, R, P, Q, \partial, T, \chi, \beta)$, no new carrier mediation role, particle persistence class, quotient comparison class, or transport witness has been introduced. Counting the presentations as physically distinct would therefore multiply descriptions without multiplying physical content. This violates quotient discipline: a representational difference that preserves all standing-bearing role data is a presentation difference, not a physical difference. $\square$

The proposition is the positive seed of gauge equivalence. It does not yet identify all such presentations. It says that physical multiplicity cannot be inferred from presentation multiplicity alone.

5 Arbitrary identification fails

The opposite error is equally serious. If presentation differences are simply declared irrelevant without boundary fixation, then gauge equivalence becomes arbitrary convention or representative selection.

Lemma 5.1 (Arbitrary presentation identification collapse). *If two local presentations are identified without preserving boundary trace, carrier mediation, particle persistence-class identity, quotient comparison, and bounded transport role, then the identification either collapses distinct standing-bearing content or imports a hidden representative selector.*

Proof. Suppose ρ_U and ρ'_U are identified without preserving the carrier, particle, quotient, boundary, transport, and classification role data. If some component differs in a standing-bearing way, the identification erases a physical distinction. If no standing-bearing difference is checked, the identification rests on a default choice of representative or convention outside the admissible boundary structure. In either case the identification fails to preserve fixed-domain admissibility. Thus presentation equivalence must be fixed by boundary-trace-preserving role data, not asserted by convention. $\square$

This is the anti-conventionalist side of the result. Gauge is not the statement that labels are arbitrary. Gauge is the admissible discipline under which presentation variation preserves physical role.

6 Boundary-trace fixation of presentation equivalence

Boundary traces provide the role anchors for admissible presentation equivalence. Without them, same-role claims become hidden representative choices.

Proposition 6.1 (Boundary-trace fixation of presentation equivalence). *If two local presentations are gauge-equivalent, their equivalence must be fixed by admissible boundary traces:*

$$\rho_U \sim_G \rho'_U \Rightarrow \text{BoundaryFix}_D(\rho_U, \rho'_U).$$

The boundary fixation preserves carrier mediation, particle persistence-class identity, quotient comparison, transport compatibility, and boundary/classification witnesses.

Proof. Gauge-equivalent presentations must preserve the same standing-bearing role. In the realized regime, sameness of role cannot be supplied by a hidden selector, arbitrary convention, or notation-only identification. It must be witnessed by admissible boundary traces that fix the carrier and particle content being preserved. If no such boundary trace is available, the alleged equivalence is either underdetermined or externally selected. Therefore admissible presentation equivalence requires boundary-trace fixation. $\square$

The proposition is the hinge of the paper. Boundary fixation prevents two complementary failures: treating equivalent presentations as different physical content, and treating non-equivalent presentations as identical by convention.

Proposition 6.2 (Gauge mediation). *If $\rho_U \sim_G \rho'_U$, then admissible modification of either presentation changes the same carrier-mediated transport/restoration organization. Thus gauge equivalence is not descriptive equivalence alone; it preserves the operative standing-bearing content carried by Φ_D and identified by Π_D .*

Proof. Same-role presentation equivalence preserves carrier mediation, particle persistence-class identity, quotient comparison, boundary trace, transport interface, and transport/restoration profile. If modifying one presentation altered a different transport/restoration organization from modifying the other, the presentations would not preserve the same standing-bearing role. Conversely, if the admissible modifications preserve the same organization, the presentations differ only in the local manner of presentation while retaining the same operative physical content. The equivalence is therefore role-preserving and mediation-preserving, not merely a relation among descriptions. $\square$

7 Gauge-role induction

Theorem 7.1 (Gauge-role induction). *In a fixed admissible realized persistence regime with field-carrier mediation Φ_D , particle persistence classes Π_D , bounded perturbation transport, quotient comparison carrier C_D , and boundary-trace fixation, presentation multiplicity that preserves the same standing-bearing physical role induces a gauge-role structure:*

$$(\Phi_D, \Pi_D, \text{BoundTrans}_D, C_D, \text{BoundaryFix}_D) \wedge \text{PresMult}_D \wedge \text{RolePres}_D \Rightarrow \exists \mathcal{G}_D.$$

Here

$$\mathcal{G}_D = (\text{Pres}_D, \sim_G, \{g_{UV}\}, \text{Fix}_{\partial, D})$$

is the minimal boundary-fixed presentation-equivalence and transition structure preserving carrier mediation and particle persistence-class identity.

Proof. By Paper 6, physical field structure occupies the role of distributed carrier mediation Φ_D . By Paper 7, particle identity is a localized persistence class Π_D supported by that mediation. Suppose the same carrier/particle content admits multiple local presentations. Presentation

multiplicity alone cannot imply physical multiplicity, since equivalent presentations preserve the same standing-bearing content. Conversely, arbitrary identification of presentations fails unless boundary traces fix the role being preserved. Therefore admissible same-role variation must be organized by an equivalence relation $\sim_G$ on local presentations, constrained by boundary-trace fixation and operative gauge mediation. Across overlapping or adjacent local continuation domains, local presentations must transition by maps g_{UV} preserving carrier mediation, particle class, quotient comparison, and transport compatibility. The family of presentations, same-role equivalence, boundary-fixed transitions, and boundary-fixation discipline is exactly $\mathcal{G}_D$. Hence gauge-role structure is induced. $\square$

The theorem does not derive a particular Lie group. It says that once the field and particle roles exist, admissible presentation freedom requires a gauge role. Lie groups and bundles are later ways this role can be mathematically specialized.

8 Local gauge transitions

The gauge role is local because the presentations being related are local carrier presentations over continuation-neighborhood domains. The transition object is therefore tied to local boundary compatibility, not to a global choice of representative. Operationally, a transition is gauge-role preserving only when it preserves carrier mediation, perturbation-transport compatibility, particle persistence-class identity, quotient comparison, boundary trace, and local-domain role. If any of these fail, the transition is not a gauge transition but a physical role change.

Proposition 8.1 (Local gauge transition compatibility). *If $U, V \in \text{LCD}_D$ overlap or are adjacent and $g_{UV} : \rho_U \rightsquigarrow \rho_V$ is an admissible gauge transition, then g_{UV} preserves carrier mediation, particle persistence-class identity, quotient comparison, transport witness compatibility, boundary trace, and local-domain role:*

$$g_{UV} \text{ admissible} \Rightarrow (\Phi, \Pi, Q, T, \partial, \chi, \beta) \text{ preserved across } U, V.$$

Proof. A local transition that fails to preserve carrier mediation changes the field role. A transition that fails to preserve Π_D changes particle identity. A transition that fails to preserve quotient comparison or transport compatibility changes the comparison/transport structure inherited from Papers 4 and 5. A transition without boundary trace preservation is not fixed by the admissible local-domain interface. Thus each of the listed components is required for a transition to be gauge-role preserving rather than physically content-changing. $\square$

Proposition 8.2 (Gauge transition composition constraint). *For local domains U, V, W , admissible gauge transitions compose only when the intermediate boundary traces preserve the same carrier/particle role and quotient comparison. When direct and composed transitions present the same standing-bearing role, one may write*

$$g_{UW} = g_{VW} \circ_D g_{UV}.$$

No principal-bundle, fiber-bundle, cocycle, connection, or group-action structure is assumed at this stage. The transition discipline supplies only the minimal presentation-equivalence role later occupied by those stronger mathematical formalisms.

Proof. The composition $g_{VW} \circ_D g_{UV}$ is admissible only if the presentation reached over V preserves the boundary trace needed to compare the presentation over U with that over W . If the intermediate trace drifts, the composed transition may connect presentation symbols while failing to preserve the standing-bearing role. When direct and composed transitions are asserted to represent the same role-preserving redescription, boundary and quotient preservation require their agreement at the gauge-role level. This is a composition constraint, not yet a full bundle cocycle axiom. $\square$

9 Primitive and skin-only gauge structures fail

The preceding theorem derives gauge-role structure from boundary-fixed presentation equivalence. This section excludes common substitutes.

Theorem 9.1 (No primitive gauge-group structure). *Primitive gauge-group structure cannot be imported before same-role presentation equivalence and boundary-fixed local transitions are established. Gauge-group formalisms become later specializations of the presentation-equivalence role derived here. If a proposed group preserves the derived gauge role, it is a later specialization; if it does not, it is skin, selector, or hidden structure import.*

Proof. A group is physically gauge-relevant only if its transformations preserve standing-bearing carrier/particle content under admissible local presentations. If the group transformations preserve $(\Phi, \Pi, Q, T, \partial, \chi, \beta)$, their physical relevance is grounded in the derived gauge role. If they do not, the group acts on notation, a hidden representative space, or an externally imposed structure rather than on admissible physical content. Thus primitive gauge-group structure is unnecessary and insufficient at this stage. The role comes first; group formalisms are later specializations of that role. $\square$

Lemma 9.2 (Skin-only gauge collapse). *A transformation changing only notation, phase convention, chart, coordinate label, or presentation skin carries no gauge role unless it preserves standing-bearing carrier/particle content and boundary-compatible transitions.*

Proof. Skin-level transformation changes the presentation layer without constraining the standing-bearing role. Such a change may be harmless notation, but it is not gauge structure in the physical-role sense. Gauge structure requires preservation of the content mediated by Φ_D and identified by Π_D , together with the boundary and transport compatibility that make local presentation variation admissible. $\square$

Lemma 9.3 (Gauge-multiplicity collapse). *If gauge-equivalent presentations are counted as physically distinct, the regime duplicates standing-bearing content without a new boundary/classification witness. Such duplication violates quotient discipline.*

Proof. Gauge-equivalent presentations preserve the same role data. Treating them as distinct physical content introduces more physical states than the boundary/classification witnesses license. The additional multiplicity is therefore representational rather than physical. Quotient discipline requires those presentations to be identified at the gauge-role level. $\square$

Lemma 9.4 (Convention-only gauge collapse). *If gauge equivalence is merely convention without boundary-trace-fixed role preservation, it cannot preserve carrier mediation or particle persistence-class identity. Gauge is therefore not arbitrary convention; it is admissible presentation discipline.*

Proof. Convention alone can rename a representation, but it cannot establish that two local presentations preserve the same field-carrier and particle-role content. That preservation requires boundary-trace fixation and quotient-compatible transitions. A convention lacking those anchors is either skin-only notation or hidden representative choice. $\square$

10 Boundary witnesses, charges, and later gauge physics

The abstract boundary/classification witness χ introduced in the particle paper becomes important here. Gauge structure preserves presentation variation while fixing the role by which carrier and particle content remain distinguishable. In later physical specializations, that role may be occupied by charge, spin representation, weak isospin, color representation, hypercharge, flavor classification, or other invariant data. The present paper does not derive those specific quantities. It derives why such witnesses cannot be replaced by arbitrary presentation choice.

Gauge coupling parameters likewise enter downstream. Raw coupling numbers are not the primitive gauge role. They become meaningful only relative to presentation normalization, invariant compatibility, boundary fixation, and the later dynamics that identify how carrier presentations interact. The present result is prior: before a gauge coupling can be physically meaningful, there must already be a gauge role preserving content across admissible presentation variation.

11 Relation to standard gauge theory

Standard gauge theories instantiate the role derived here in stronger mathematical form. They organize local presentation multiplicity, role-preserving transitions, overlap compatibility, local equivalence structure, and transport-compatible presentation freedom. In classical and quantum field theory, gauge formalisms commonly add representation spaces, Lie groups, connection or covariant-derivative structures, curvature, dynamics, and sometimes quantized gauge bosons [15, 16, 17, 18, 19, 20]. The present theorem does not assume any of that formal machinery. It identifies the prior physical role such machinery serves.

Standard gauge ingredient	Status in this paper
Local presentation freedom	derived as PresMult_D
Gauge-equivalent descriptions	derived as $\sim_G$
Local transition functions	derived as boundary-fixed g_{UV}
Lie group / algebra	downstream specialization
Connection / curvature	downstream transport geometry
Gauge coupling values	downstream normalization and dynamics
Yang–Mills equations	downstream field dynamics
Gauge bosons / quantization	downstream particle/field specialization

This role-first ordering is essential. Gauge structure is not added as pure redundancy to an already complete field/particle description. It is forced by the requirement that multiple local presentations preserve the same field-carrier and particle-role content without arbitrary representative choice.

12 No hidden-variable gauge reading

Gauge presentations are not hidden variables. Gauge-equivalent presentations do not encode extra physical state unless boundary witnesses distinguish them. This distinction is important in nonclassical settings: boundary-level compatibility, entanglement, and nonseparable correlations do not license interpreting gauge presentation freedom as a reservoir of hidden local facts. Gauge equivalence removes presentation redundancy while preserving physical role; it does not add unobserved particle content or Bell-local hidden variables [22, 21].

13 Gauge role and bounded transport

Gauge transitions must be compatible with bounded perturbation transport. A local presentation change that preserves static carrier labels but breaks the transport witness structure is not gauge-equivalent in the realized regime. The reason is inherited from Paper 5: transport is not path syntax but witness-mediated preservation of perturbation burden, quotient comparison, restoration status, and local-domain role. Gauge transition compatibility must therefore commute with the admissible transport role whenever both structures are asserted over the same local domains.

For a transport witness $\tau_D(U, V, p)$ and a presentation transition g_{UV} , the relevant compatibility is not a full dynamical covariance law. The present paper does not yet derive a covariant derivative or connection. The minimal requirement is role preservation: the presentation transition cannot turn an admissible transport witness into an inadmissible one for the same carrier/particle content.

Proposition 13.1 (Transport compatibility of gauge transition). *If g_{UV} is an admissible gauge transition between local presentations and $\tau_D(U, V, p)$ is an admitted perturbation-transport witness for the same carrier/particle role, then applying the presentation transition preserves the admissibility class of the transport witness.*

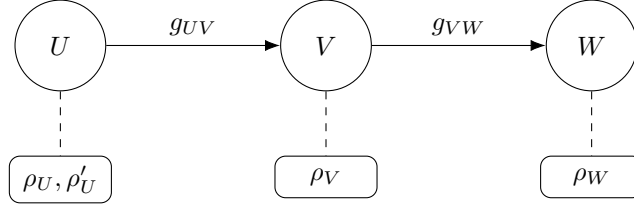
Proof. The transport witness preserves perturbation burden, boundary trace, quotient identity, restoration status, comparison class, and local-domain role. A gauge transition preserving the same carrier/particle content must preserve those same role data at the presentation level. If it did not, the transition would change the transport status of the content it claims merely to redescribe. That would be role change, not gauge equivalence. Thus transport-witness admissibility is invariant under the admitted gauge transition in the minimal role sense. $\square$

14 Finite witness model

A finite witness model illustrates how the theorem functions without defining gauge theory as graph data. Let

$$G_D = (V, E, \Phi, \Pi, \text{Pres}, g)$$

where V is a finite family of local continuation domains, E records overlap or adjacency, $\Phi(v)$ is the carrier profile at v , $\Pi(v)$ is the particle persistence class supported there when present, $\text{Pres}(v)$ is the family of admissible local presentations, and g_{uv} is a candidate transition between presentations.



boundary-fixed presentation transitions preserve $\Phi, \Pi, Q, T, \partial, \chi, \beta$

The relevant failure modes are as follows:

1. If $\text{Pres}(v)$ has no multiplicity, no local gauge-role transition is required at v .
2. If presentation multiplicity is counted as physical multiplicity, quotient discipline is violated.
3. If presentations are identified arbitrarily, a selector is imported.
4. If g_{uv} lacks boundary trace fixation, carrier or particle role drifts.
5. If a primitive group is imported before role preservation, hidden structure is introduced.
6. If boundary-fixed presentation equivalence and compatible transitions are supplied, the gauge role is instantiated.

The witness graph instantiates one admissible realization of gauge-role structure. It does not define general gauge theory as graph data.

15 Gauge-source route exhaustion

The derivation can be stated as an exhaustion of possible sources of gauge structure.

Theorem 15.1 (Gauge-source route exhaustion). *Any proposed source of gauge structure in a fixed admissible realized persistence regime is one of the following broad classes:*

1. *boundary-fixed presentation equivalence;*
2. *arbitrary convention;*
3. *primitive gauge-group structure;*
4. *physical multiplicity of presentations;*
5. *hidden or skin-only presentation structure;*
6. *transport-incompatible transition structure.*

Routes 2–6 fail to preserve fixed-domain admissibility unless they reduce to route 1. Therefore the surviving gauge source is boundary-fixed presentation equivalence.

Proof. Arbitrary convention lacks boundary-fixation authority. Primitive gauge-group structure is physically relevant only if it preserves the derived presentation-equivalence role; otherwise it is hidden or skin-level. Counting presentations as physical multiplicity duplicates content without a boundary witness. Hidden or skin-only presentation structure either imports a selector or carries no standing-bearing role. Transport-incompatible transitions fail to preserve the carrier/particle content whose redescription they claim to relate. Each failing route either collapses physical content, imports hidden structure, or reduces to boundary-fixed preservation of carrier/particle role. Hence route 1 is the only surviving source. $\square$

16 Normal form

The result can be summarized as

$$(\Phi_D, \Pi_D, \text{BoundTrans}_D, C_D, \text{BoundaryFix}_D) \wedge \text{PresMult}_D \wedge \text{RolePres}_D \Rightarrow \text{GaugeRole}_D$$

with

$$\text{GaugeRole}_D \equiv \mathcal{G}_D = (\text{Pres}_D, \sim_G, \{g_{UV}\}, \text{Fix}_{\partial, D}).$$

Gauge structure is therefore the minimal admissible presentation-equivalence and local-transition discipline preserving field-carrier mediation and particle persistence-class identity.

17 Conclusion

Paper 1 established that realized persistence cannot be physically null. Paper 2 established bounded standing-localization as the origin of physical locality. Paper 3 derived continuation-neighborhood domains. Paper 4 derived the metric role of quotient-stable comparison. Paper 5 derived bounded perturbation transport as the finite-propagation constraint role. Paper 6 derived fields as persistence carriers. Paper 7 derived particles as localized persistence classes. The present paper shows that once field-carrier and particle-role content admit multiple local presentations, admissibility requires boundary-trace-fixed presentation equivalence. What has been isolated is the minimal presentation-equivalence and transition structure required to preserve carrier-mediated particle content across local continuation domains. Gauge structure is therefore neither primitive redundancy nor arbitrary convention; familiar gauge-theoretic structures become later specializations of this role. Lie groups, connections, curvature, Yang–Mills dynamics, coupling values, gauge bosons, and quantized gauge fields are downstream developments of the boundary-fixed gauge role derived here.

References

- [1] A. J. Maley. *Realized Persistence Cannot Be Physically Null*. Unpublished manuscript, 2026.
- [2] A. J. Maley. *Standing Localization and the Origin of Physical Locality*. Unpublished manuscript, 2026.
- [3] A. J. Maley. *Continuation Loci and the Origin of Local Domains*. Unpublished manuscript, 2026.
- [4] A. J. Maley. *The Metric Role from Continuation Comparison*. Unpublished manuscript, 2026.
- [5] A. J. Maley. *Bounded Perturbation Transport and the Origin of Finite-Propagation Constraint*. Unpublished manuscript, 2026.
- [6] A. J. Maley. *Fields as Persistence Carriers*. Unpublished manuscript, 2026.
- [7] A. J. Maley. *Particles as Localized Persistence Classes*. Unpublished manuscript, 2026.
- [8] A. J. Maley. *Boundary-Trace Fixation of Continuation Loci under AASC*. Unpublished manuscript, 2026.

- [9] A. J. Maley. *Gauge Coupling Parameters and Forced Uniqueness*. Unpublished manuscript, 2026.
- [10] A. J. Maley. *A Classification Theorem for Admissible Standard Model Structures*. Unpublished manuscript, 2026.
- [11] A. J. Maley. *Structura Ex Necessitate Standardis Modelis*. Unpublished manuscript, 2026.
- [12] A. J. Maley. *The AMetric Boundary as a Unique Regime-Invariant Constraint on Admissibility-Bearing Construction*. Unpublished manuscript, 2026.
- [13] A. J. Maley. *Anchor, Tensor, and Skin: A Fixed-Domain Decomposition Theorem for Admissible Description*. Unpublished manuscript, 2026.
- [14] A. J. Maley. *Quantum Entanglement as Boundary-Level Compatibility*. Unpublished manuscript, 2026.
- [15] H. Weyl. Elektron und Gravitation. I. *Zeitschrift fuer Physik*, 56:330–352, 1929.
- [16] C. N. Yang and R. L. Mills. Conservation of isotopic spin and isotopic gauge invariance. *Physical Review*, 96:191–195, 1954.
- [17] R. Utiyama. Invariant theoretical interpretation of interaction. *Physical Review*, 101:1597–1607, 1956.
- [18] M. Nakahara. *Geometry, Topology and Physics*. Institute of Physics Publishing, 2nd edition, 2003.
- [19] S. Weinberg. *The Quantum Theory of Fields, Volume II: Modern Applications*. Cambridge University Press, 1996.
- [20] M. E. Peskin and D. V. Schroeder. *An Introduction to Quantum Field Theory*. Westview Press, 1995.
- [21] R. Haag. *Local Quantum Physics: Fields, Particles, Algebras*. Springer, 2nd edition, 1996.
- [22] J. S. Bell. On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika*, 1:195–200, 1964.